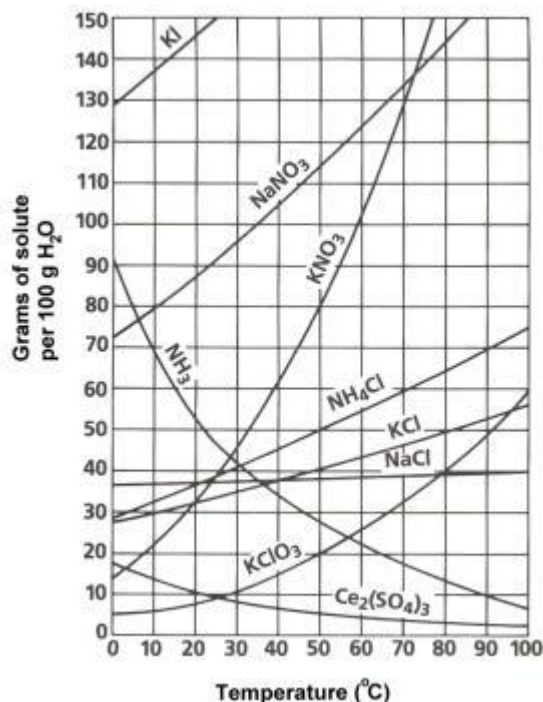


Solubility Curves

There are charts and tables available that we can use to get an idea of how soluble a certain solute is in a certain solvent. We will take a look at two of them in these next two sections.

Solubility curves, like the one shown here, tell us what mass of solute will dissolve in 100g (or 100mL) of water over a range of temperatures.

You'll notice that for most substances, solubility increases as temperature increases. In solutions involving liquids and solids typically more solute can be dissolved at higher temperatures. Can you find any exceptions on the chart?



Here's an example of reading the chart. Find the curve for KClO₃.

At 30°C approximately 10g of KClO₃ will dissolve in 100g of water. If the temperature is increased to 80°C, approximately 40g of the substance will dissolve in 100g (or 100mL) of water.

Here are some for you to try.

What mass of solute will dissolve in 100mL of water at the following temperatures. Also determine which of the three substances is most soluble in water at 15°C.

1. KNO₃ at 70°C
2. NaCl at 100°C
3. NH₄Cl at 90°C

On a solubility curve, the lines indicate the concentration of a **saturated solution** - the maximum amount of solute that will dissolve at that specific temperature. The molar concentration of the substance can be calculated, as shown by this example:

Determine the molarity of a saturated NaCl solution at 25°C.

Solution

We can see from the curve that about 38 g of NaCl dissolves in 100mL at 25°C. Molarity, M, has the units mol/L, so we want to convert 38g /100mL to mol/L. Since we need to convert from grams to moles, we will need to use the molar mass of NaCl, which is 58.5 g/mol

desired unit		molar mass		answer
$\frac{\text{mol}}{\text{L}}$	$= \frac{38\text{g}}{\text{L}}$	$\times \frac{1 \text{ mol}}{58.5\text{g}}$	$= \frac{6.5 \text{ mol}}{\text{L}}$	OR 6.5M
				answer

Values on the graph below a curve represent **unsaturated solutions** - more solute could be dissolved at that temperature. Values above a curve represent **supersaturated solutions**, a solution which holds more solute that can normally dissolve in that volume of solvent.

Some examples:

What term - saturated, unsaturated, or supersaturated - best describes:

- a solution that contains 70g of NaNO₃ per 100 mL H₂O at 30°C
- a solution that contains 60g of dissolved KCl per 100 mL H₂O at 80°C

Practice Questions

1. When a solvent holds as much of a solute as it normally can at a given temperature we say the solution is _____?
2. When a solvent has less dissolved solute than it normally can at a given temperature we say the solution is _____?
3. Graphs with temperature on the x-axis and solubility of a solute in water on the y-axis have plotted lines called solubility curves. Each point exactly on the solubility curve represents a solution that is _____?
4. Any data point beneath a solubility curve represents a solution that is _____?
5. Solids dissolved in water generally are _____ (more? or less?) soluble as temperature rises.
6. If you cool a solution that was saturated and crystal solute does not form in the solution, then the solution must have become _____?

7. Gases dissolved in water generally are _____(more? or less?) soluble as temperature rises.
8. Using the solubility graph on your reference tables, express the solubility (with units - for example, 80 g/100 g water) of KNO_3 at 40°C .
9. How many grams of KNO_3 will dissolve in 50 grams of water at 40°C ? (Hint: first find how many grams of KNO_3 will dissolve in 100 grams of water from the solubility graph, then set up a proportion.)
10. How many grams of water will be needed to dissolve 25 grams of KNO_3 at 40°C ? (Hint: same hint as in #11)
11. If 8 mg of CO_2 will dissolve in 1000 grams of water at 10°C (saturated solution), how many mg of CO_2 can be dissolved in a can of soda (353 grams of water) at the same cool temperature?
12. If the carbonated beverage in Question #11 were allowed to warm to room temperature but none of the carbon dioxide escaped (can is still sealed), what term would we use to describe this solution?
13. If the can from Question #12 were finally opened at room temperature and quickly poured into a glass, what would you observe in the solution? (crystals forming? a rainbow of colors? bubbles? ice cubes forming?)